AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE399 – Project Phase 1 (Technical Progress Report - Review 2)
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Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23547	Sheela Akshar Sakhi	Dr. G R RAMYA, Dr. VANDHANA S Dept. of CSE	Type 1 Research	.../ 50
CB.SC.U4CSE23529	Hasini Reddy M			
CB.SC.U4CSE23761	Kousik Sarma Lakkaraju			
CB.SC.U4CSE23363	Haswitha K			
CB.SC.U4CSE23753	V.Chakravarthy			

1 Project Proposal and objectives

Overall Project Idea: To design a fully decentralized, privacy-preserving Federated Personalized Drift-Aware Attention Framework (FPDAF) specifically tailored for multivariate healthcare time-series analysis (e.g., ICU patient risk monitoring and forecasting).

Objectives to Achieve:

1. Enable cross-hospital predictive modeling on patient data without violating HIPAA/GDPR privacy constraints.
2. Manage high patient heterogeneity across hospitals using explicit Local Personalization Heads.
3. Detect when a patient's core physiological state changes using CUSUM Drift Detection algorithms to automatically retrain and dynamically adapt models locally.
4. Integrate Temporal Attention weights to provide clinicians with transparent insights into exactly what historical vital signs caused a risk prediction.

2 Introduction and Background

Continuous monitoring systems in Intensive Care Units (ICUs) collect immense streams of physiological data (HR, BP, SpO2). Early detection of patient deterioration using Deep Learning models can prevent fatalities like Sepsis or Cardiac Arrest. However, massive clinical models require centralized data, violating privacy laws. While Federated Learning (e.g., FedAvg) resolves the data-sharing issue, it fails to account for non-IID data distributions (how demographics and hospital protocols vary significantly between facilities).

Our literature review explored standard Federated methods. FedAvg (McMahan 2017) trains efficiently but lacks personalization. FedPer (Arivazhagan 2020) personalizes models to local distributions but lacks any mathematical mechanism for tracking concept drift over time. FedXAI works (Kalakoti 2025) focus mostly on classification datasets, not complex high-frequency critical care time-series. Thus, a significant gap exists in achieving privacy, personalization, drift monitoring, and clinical explainability simultaneously.

3 Execution plan

The subdomains required to solve this problem include Data Engineering (PhysioNet Challenge 2019 dataset ingestion, structuring temporal windows), Deep Learning Architecture (LSTM backbone

and Softmax attention mechanics), Decentralized Systems (FedAvg aggregation protocols), Model Customization (extracting independent local decision heads), and Bayesian Statistics/Math (using CUSUM residual monitoring).

Our plan mapped these subdomains to execution steps: initially solidifying the theoretical bounds and mathematical triggers in Phase 1, followed by developing the PyTorch dataset pipeline, establishing the generic FedAvg baseline model locally, and finally attaching our novel drift-detector and attention overlays in Phase 2.

4 Fundamentals which you have learned in the semester

To prepare for implementation, the team completed an extensive curriculum covering the fundamental requirements:

- Textbooks & Papers: Thoroughly studied the foundational architectures of Federated Learning (FedAvg, pFL, and FedPer) to mathematically understand secure aggregation.
- PyTorch Engineering: Used MOOCs and official documentation to learn how to explicitly sever Neural Network graphs in PyTorch (allowing us to separate "Global Backbones" from isolated "Local Heads") and simulate custom training loops iteratively without data centralization.
- Statistical Mechanics: Learned the equations and thresholds defining the Cumulative Sum (CUSUM) drift detection standard to track temporal shifts in residual streams.

5 Architecture diagram and explanation of the design

The FPDAF architecture is a split-layer design. The Global Temporal Backbone consists of the LSTM layers responsible for learning generalized physiological laws across all participating hospitals.

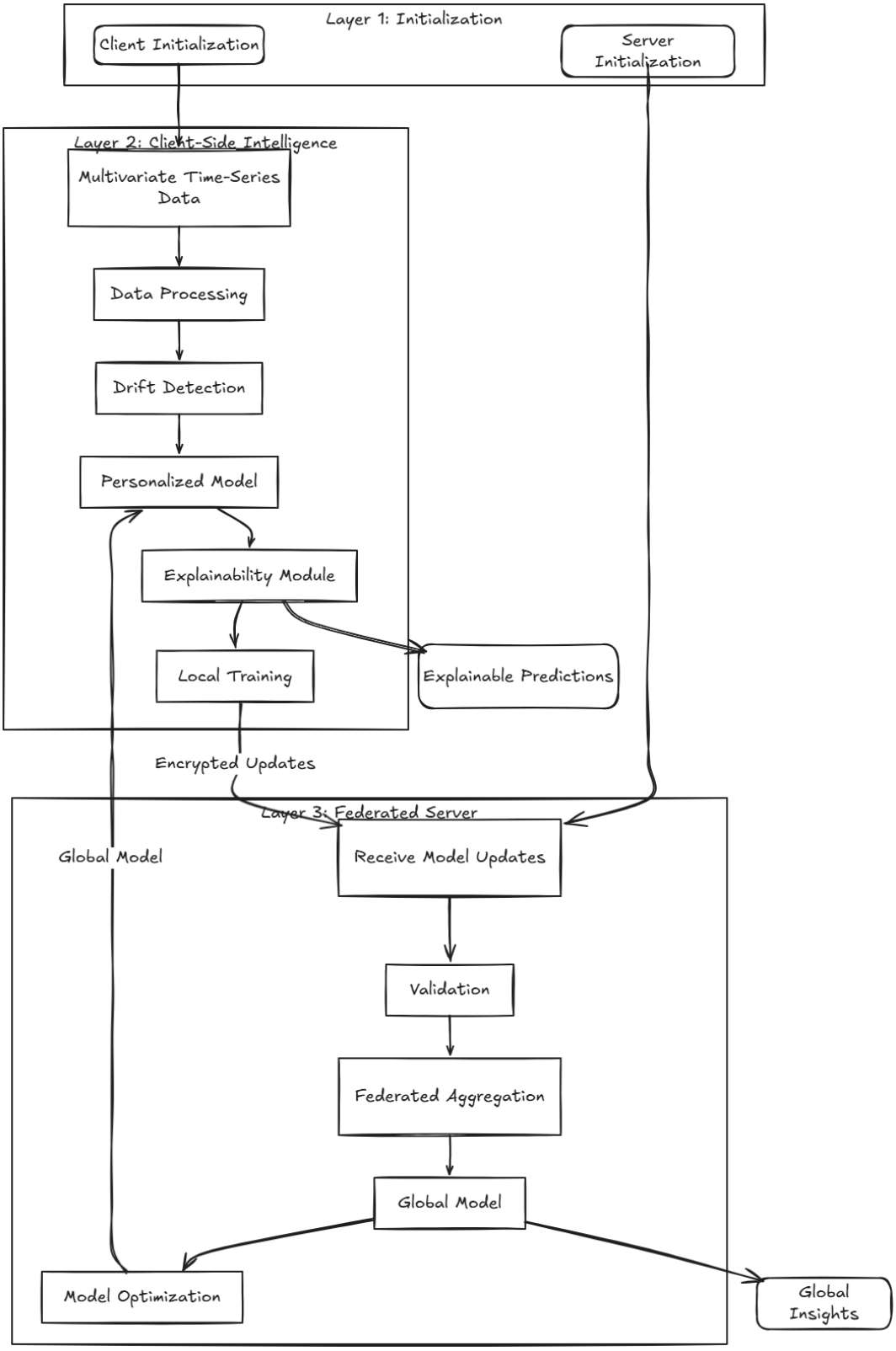

Figure 1: System Architecture highlighting separation of Global Backbone (server-synced) and Local Heads (hospital-isolated).

The Attention Layer sits atop the backbone to compute temporal context vectors based on recent physiological spikes, giving transparent x_i timestamps feature importance. Finally, the Personalization Head is isolated completely to the hospital node; it never transfers its gradients to the server, ensuring the final clinical decision is strictly correlated to the local hospital's demographic

distribution.

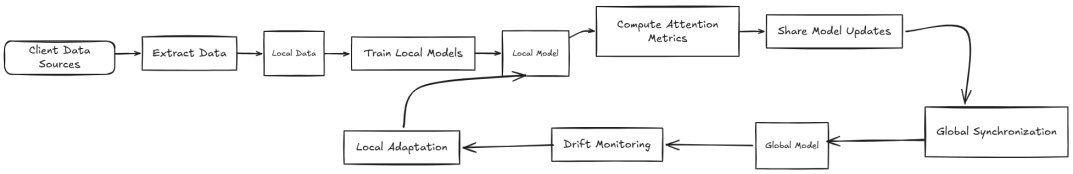

Figure 2: FPDAF System Dataflow Mapping - Illustrating end-to-end data processing.

6 Summary of works completed / research findings (Technical Evidence)

During this semester (Phase 1), the following milestones were completed mapping to our Technical Validation & Evidence requirements:

1. Domain Consolidation: Successfully shifted use-case entirely into ICU Healthcare datasets.
2. Mathematical Definition: Derived the CUSUM threshold trigger mechanism $S_t = \max(0, S_{t-1} + (e_t - \mu - k))$ specifically for linking to local model resets.
3. Architecture Creation: Finalized the fully decentralized, visually mapped FPDAF blueprint.
4. PyTorch Baseline Simulation Setup: Created the GitHub repository (https://github.com/aksharsakhi/23CSE399-Project-Phase-1), synced a foundational FedAvg-PyTorch implementation baseline, mocked a SimulatedICUDataset class, and began structurally coding the decoupled model heads in Python.

7 Plans for the oncoming semesters

Having successfully acquired the PhysioNet Challenge 2019 (Sepsis Prediction) open-access dataset, Phase 2 will aggressively focus on data preprocessing, missing value imputation, and constructing sequential time-windows. The team will establish the foundational generic FedAvg baseline locally, and subsequently train the custom decoupled FPDAF pipeline across dynamically simulated hospital nodes using PyTorch. Finally, rigorous benchmarking will be conducted, measuring drift recovery delays alongside formal evaluation metrics including Accuracy, Precision, Recall, F1-Score, and AUC-ROC to definitively quantify the optimization & personalization gains.

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